

Blind Frames, Broken Stories: the Verification Bottleneck in Scientific Video Generation

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Abstract

Turning an academic paper into a short educational video is now within reach of generative AI: an LLM orchestrates the process, text-to-image and image-to-video models supply the visuals, and speech synthesis provides the narration. One step in this chain is far harder than the rest. Before a figure is animated, someone has to check that it is actually correct, because a video diffusion model renders a wrong diagram with exactly the same fidelity as a right one. We address this with the Agentic Director Pipeline, a multi-stage system that produces 45-75 second scientific videos behind explicit factual and visual verification gates. To measure how hard the verification step really is, we built a diagnostic corpus of 70 scientific figures with controlled error injections and tested five current VLMs as verifiers: GPT-5.1, Claude Sonnet 4, Gemini 2.5 Pro, Qwen3.6-VL, and DeepSeek-VL2. Neither zero-shot nor Chain-of-Thought prompting holds up; in the worst configurations both the over-rejection rate and the miss rate climb to 0.86. Three recurring behaviors account for the failures: Knowledge Contamination, Spatial Insensitivity, and the Chain-of-Thought Paradox. Together, they motivate a verifier built around grounded structural schemas and an adversarial Advocate-Critic check.

1. Introduction

Researchers no longer reach their audience through journals alone. A growing share of scientific communication now travels through digital media (Brossard & Scheufele, 2013),

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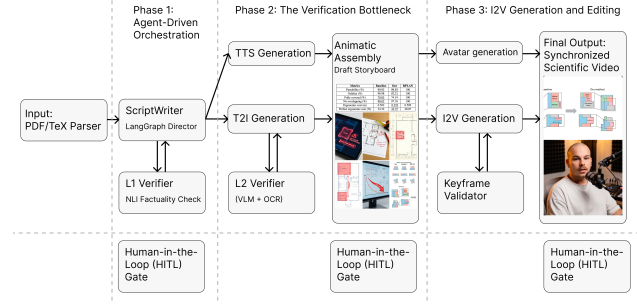


Figure 1. The complete Agentic Director Pipeline. The system takes in PDF documents, uses graph-based agents to orchestrate the script and the generation prompts, checks textual factuality against the source (L1 Verifier), validates the visual assets with VLMs (L2 Verifier), and finally renders the verified assets into a synchronized video through diffusion models.

and video platforms push that reach further still (Welbourne & Grant, 2016). The attention this attracts on social media has even started to feed back into how scholarly impact is measured (Sugimoto & Thelwall, 2017). Building a video from a PDF, meanwhile, has become technically routine: LLM-based agents and multimodal generation make the whole pipeline feasible end to end (Wu et al., 2023). What that automation quietly drops is the editorial judgment that, in the manual process, stands between a peer-reviewed figure and the animation an audience finally sees.

The risk here is asymmetric. A video diffusion model reproduces an incorrect diagram just as faithfully as a correct one (Ho et al., 2022; Blattmann et al., 2023; Brooks et al., 2024), so any error sitting in a keyframe before I2V synthesis spreads into every frame that follows and yields a fluent video that happens to be wrong. Checking the keyframe before it enters the generator is therefore not a refinement but a safety gate the rest of the pipeline depends on. A VLM is the obvious candidate for that gate, yet it is far from clear one can be trusted with it. These models hallucinate objects that were never in the image (Liu et al., 2023), and they lean on language priors instead of the pixels in front of them (Li et al., 2023). Many of the benchmarks that make them look dependable, it turns out, can be solved from text alone (Chen et al., 2024). The weakness bites hardest exactly where this task lives: Tong et al. (Tong et al., 2024)

trace it to fine-grained spatial judgment. And a scientific figure is made of such judgments. Whether an arrow runs the right way, whether an axis is scaled correctly, whether a block sits where it belongs – none of it survives in the surface text a model can fall back on.

Our main contribution is an empirical picture of this bottleneck. We assemble a diagnostic corpus of 70 scientific figures with controlled error injections, evaluate five current VLMs under both zero-shot and Chain-of-Thought prompting, and isolate three failure patterns that recur whether the model is proprietary or open-weight. The diagnosis then drives a design: the Agentic Director Pipeline, a multi-stage PDF-to-video system in which a grounded Advocate-Critic loop acts as the L2 verification gate ahead of image-to-video synthesis.

2. Agentic Director Pipeline

The pipeline runs in three stages – Agent-Driven Orchestration (Phase 1), Visual-Semantic Verification (Phase 2), and I2V Generation (Phase 3). Output moves forward only once it has been verified; a failed check sends it back into a regeneration loop before anything advances.

Phase 1: Agent-Driven Orchestration. A LangGraph-based writer-agent (LangChain Inc., 2024) – LangGraph being a framework for stateful, multi-actor LLM applications – sits at the centre and drives the whole process from a single scientific paper. A publication parser first pulls out the text and the original graphics. GPT-4o then writes a script tuned for visual dynamics and marks up the segment timelines. The Director also reads each figure’s caption and the text around it, and from that builds a structural schema: the nodes it expects, the edges between them, and their labels. When a concept has no direct visual form, the agent writes a descriptive prompt and asks a T2I model for a keyframe instead.

Factuality of the text is enforced by Natural Language Inference (NLI) against the source article. A DeBERTa-based NLI model (He et al., 2021) matches every script statement to the passages most relevant to it and flags anything that contradicts the source or is left unsupported, sending it back for regeneration. This is the L1 Verifier, and its job is narrow: keep the LLM from asserting facts the paper never states.

Phase 2: Visual-Semantic Verification. Before the expensive rendering step begins, the system stitches together a draft animatic – TTS voiceover, a digital avatar, and the starting keyframes. Verification then happens on two levels. The L2 Verifier inspects each starting keyframe before I2V generation; the Keyframe Verifier watches frame-level quality once the video has been rendered.

L2 Verifier: Starting Frame Verification. Where L1 works on text through NLI, the L2 Verifier looks at the visual structure itself – the topology of the graph, the spatial layout, and whether text and image agree. Each keyframe arrives

bundled with three things: the script statement, a figure-type tag (produced by a separate classification prompt sent to the same VLM), and the structural schema the Director prepared. The tag decides how the frame is checked. Architectures get graph tracing, charts get axis-and-value extraction, and pipelines get flow-consistency checking. The schema, in turn, lists nodes and directed edges for an architecture or a pipeline, or axes, series, labels, and value relations for a chart.

A two-agent loop implements this check. The *Advocate* argues that the figure is consistent with the statement; the *Critic* starts from the opposite assumption – that exactly one error is present – and tries to pin it down. Each returns a verdict, an error type, and a pointer to where the problem lies. When both pass, the frame goes to Phase 3. When both fail, their two explanations go to the Director, who diagnoses the issue, corrects the prompt, and triggers T2I regeneration. A disagreement is escalated to the Director; if it is still unresolved, or a second round again ends in FAIL, the case goes to a human operator.

Keyframe Verifier: Frame-Level Video Monitoring. Diffusion clips can drift in meaning as they play out, so the system samples one frame every 0.5-1 second and checks it against two references at once: the current script statement and the Director’s scene description. Grounding the frame in both lets the system tell harmless visual variation apart from a real departure from the narrative. Two agents look at every sampled frame. The *Critic Agent* leans toward rejection, and the *Advocate Agent* pushes back on what it finds – an asymmetry that absorbs the over-rejection pattern reported in Section 3 without softening the standard. Two passes accept the frame; two failures send the Director back to revise the I2V prompt and regenerate the 2-4 second clip in question. As before, disagreements go to the Director, and anything still unresolved after the second round goes to a human.

Phase 3: I2V Generation and Editing. The verified starting frames pass to I2V diffusion models, each producing a 2-4 second clip. A final editing module lines the clips up with the TTS track and assembles them into a coherent 45-75 second video. Since every starting frame has already cleared the L2 gate, the model is animating content that is known to be correct – so temporal consistency now works for the pipeline instead of against it.

3. Evaluation Results

3.1. Experimental Design

To characterise the bottleneck that any such pipeline has to clear, we evaluate the visual verifiers on their own. The diagnostic corpus holds 70 scientific figures drawn from three areas – neural-network architectures, evaluation metrics, and system pipelines – and we inject a targeted error into half of them. That leaves 35 correct figures and 35 hand-modified ones, each carrying a single injected error,

which gives unambiguous ground truth. The figures come from three sources: architectural diagrams from Sebastian Raschka’s LLM Architecture Gallery; benchmark charts taken from the Llama 3 and DeepSeek-VL2 (DeepSeek-AI, 2024) papers, among other landmark works; and system schematics, including the canonical diagrams of *Attention Is All You Need* (Vaswani et al., 2017), CLIP (Radford et al., 2021), ResNet (He et al., 2016), and RLHF (Ouyang et al., 2022).

Every injected modification falls into one of three classes. A structural error breaks the topology – a block displaced or swapped with its neighbour, a component deleted, a superfluous one added, or an arrow pointing the wrong way. A numerical error leaves the topology alone but corrupts a value: the embedding dimensionality, the layer or attention-head count, the vocabulary size, or parameters borrowed from the wrong model. A semantic error keeps topology and numbers intact and attacks the labels instead – a component renamed, attribution flipped between two models, or an abbreviation swapped for the wrong one.

We evaluate five leading multimodal models, all with built-in OCR. Three are proprietary – Claude Sonnet 4 (Anthropic, 2025), GPT-5.1 (OpenAI, 2025), and Gemini 2.5 Pro (Google DeepMind, 2025) – and two are open-weight or openly documented: Qwen3.6-VL (Qwen Team, 2026) and DeepSeek-VL2 (DeepSeek-AI, 2024). Each is tested under two prompting strategies. The zero-shot prompt says almost nothing: decide whether an error is present, answer PASS or FAIL. The Chain-of-Thought prompt instead walks the model through it step by step – name the figure type, list every visible element, trace the data flow or read off the chart values, check each element against what it knows, and only then commit to a verdict.

In this setup the model sees only the image, with no reference description attached – a deliberate choice that isolates its capacity for visual-structural reasoning alone. The choice surfaces a failure mode that matters in practice: stripped of a reference, models fall back on parametric memory rather than analysing what is in front of them. We report ordinary accuracy, $\text{Acc} = (TP + TN) / (TP + TN + FP + FN)$, together with TPR, TNR, the Over-rejection Rate ($\text{OR} = 1 - \text{TNR}$), and Error Localization Accuracy (LocAcc) – the share of FAIL verdicts that correctly name and place the injected error.

3.2. Results

No configuration reached operational reliability. The strongest result – Gemini 2.5 Pro with CoT, at $\text{Acc} = 0.71$ – still sits well under what autonomous operation would require, and the open-weight models fail in much the same way, which points to a limitation that is fundamental rather than a matter of scale.

Table 1. Performance of 5 SOTA VLMs on scientific figure verification (70 figures). Acc is ordinary accuracy over all examples, $\text{OR} = 1 - \text{TNR}$, and LocAcc is the fraction of FAIL verdicts with correct error localization.

MODEL $\times$ PROMPT	ACC	TPR	TNR	OR	LOCACC
CLAUDE S. 4 — ZERO	0.54	0.86	0.21	0.79	0.50
CLAUDE S. 4 — CoT	0.64	0.43	0.86	0.14	0.67
GPT-5.1 — ZERO	0.43	0.14	0.71	0.29	0.00
GPT-5.1 — CoT	0.50	0.71	0.29	0.71	0.40
DEEPSEEK-VL2 — ZERO	0.47	0.50	0.44	0.56	0.42
DEEPSEEK-VL2 — CoT	0.60	0.78	0.43	0.57	0.53
QWEN3.6-VL — ZERO	0.50	0.86	0.14	0.86	0.65
QWEN3.6-VL — CoT	0.67	0.64	0.71	0.29	0.67
GEMINI 2.5 PRO — ZERO	0.51	0.86	0.17	0.83	0.83
GEMINI 2.5 PRO — CoT	0.71	0.86	0.57	0.43	0.67

Zero-shot prompting is dominated by a strong over-rejection bias. Claude Sonnet 4, Qwen3.6-VL, and Gemini 2.5 Pro throw out correct figures at rates of 0.79, 0.86, and 0.83 – inventing errors that are not there, and doing so confidently enough to rule out autonomous use. CoT prompting pulls the over-rejection down, but the cost is missed errors: Claude Sonnet 4 under CoT lets 57% of the modified figures slip past, and GPT-5.1 zero-shot waves through 86% of injected errors outright. The localization is worse still. Even when a FAIL verdict is correct, it often points to the wrong place – GPT-5.1 zero-shot scores $\text{LocAcc} = 0.00$, never once localizing a FAIL correctly across the whole corpus.

Table 2 splits the failures by injected error type, drawing on the experiment-tracker annotations and projecting the observed proportions onto the full set of 35 modified figures. The result lines up with the mean TPR in Table 1: weighted by the error-type counts, the average detection rate comes to 0.66 up to rounding. Most of the projected failure mass is structural – a typical verifier would miss around 9 structural cases against just 2 numerical and 2 semantic ones. The weak point of the verification stage, then, is reasoning about topology and layout, not OCR or surface labels on their own.

Table 2. Detection and localization performance by error type across all model configurations. Detection = TP rate (fraction of modified figures correctly flagged as FAIL). Miss = $1 - \text{Detection}$. LocAcc = fraction of correct FAIL verdicts with correct error localization.

ERROR TYPE	SAMPLE n	DETECTION	MISS	LOCACC
STRUCTURAL	25	0.63	0.38	0.40
NUMERICAL	5	0.67	0.33	0.63
SEMANTIC	5	0.83	0.17	0.80

The pipeline is built around that finding. The Director hands the L2 Verifier a compact structural schema alongside each image, and the verifier is told to confirm the expected nodes, edges, component roles, and arrow directions before it commits to a verdict.

3.3. Discussion

A closer look turns up three behaviours that recur across both proprietary and open-weight architectures.

Knowledge Contamination. With no reference to work from, the models fall back on what they already know. The moment a recognisable identifier shows up in a diagram, the model checks it against its training data instead of reading the figure’s actual structure. On a correct DeepSeek-V3 benchmark chart, for instance, GPT-5.1 returns FAIL because the reported numbers clash with the public results it remembers – mistaking a visual-verification task for fact-checking. The lesson generalises: any verifier that leans on model knowledge alone, without a grounded reference, will keep making that same substitution.

The Advocate-Critic design pushes back on this directly. The Critic is told to hunt for inconsistencies inside the figure rather than weigh it against parametric knowledge, while the Advocate ties the figure to the supplied script statement as a local ground truth. With a concrete reference in hand, the model has far less reason to reach for external priors.

Spatial Insensitivity. The models read the labels but cannot follow the directed topology. Take a modified CLIP diagram in which the image- and text-encoder inputs are swapped (Figure 4 shows an analogous structural swap): Claude Sonnet 4 zero-shot lists every visible label correctly and then issues PASS, never noticing that the cross-modal flow is broken. A modified OLMo 2 architecture with the RMSNorm and Attention blocks transposed produces the same outcome (cf. Figure 3 for a block swap) – several models check the text labels against memory, declare the figure correct, and miss the structural violation entirely. This echoes recent evidence on the spatial shortcomings of current VLMs (Tong et al., 2024).

Chain-of-Thought Paradox. Here CoT works less like a systematic check and more like a way of talking itself into an answer. On a modified GPT-2 XL diagram (Figure 2 shows a comparable component deletion), Claude Sonnet 4 with CoT pulls out all the labels, validates each one against memory, and reasons its way step by step into believing the figure is fine – then issues PASS on a figure that carries an injected structural defect. Zero-shot mode does flag the same figure, but for the wrong reason. Neither mode is calibrated.

Each of these gets its own answer in the design. By making the verifier commit to a domain-appropriate strategy before it ever looks at the image, the figure-type classifier blunts Knowledge Contamination. The Advocate-Critic loop hands one agent the explicit job of tracing topological consistency, which is aimed squarely at Spatial Insensitivity. And fixed prompt templates hold the CoT process to a set protocol, leaving no room for the open-ended self-persuasion seen above.

For long-horizon video the consequence is plain. If a VLM cannot reliably verify a single keyframe, then every down-

stream claim about temporal consistency is resting on an assumption nobody has checked. A bigger general-purpose VLM is unlikely to fix that. What the task calls for is a domain-specific verifier trained to treat spatial topology as a primary signal rather than a by-product of reading labels – and the thing standing in the way is the shortage of specialised training data. The 70 annotated figures here are a proof of concept; scaling the benchmark up is the obvious next step. The scope stops at the AI/ML domain, too, and carrying the taxonomy into fields like biotechnology or chemistry will mean building error categories suited to them.

4. Limitations

The diagnostic corpus is small on purpose (70 figures) and confined to AI/ML publications, which limits how far the observed failure rates generalise. Its error-type mix is also deliberately uneven – 25 structural errors against 5 numerical and 5 semantic – chosen to reflect how often topology-level mistakes actually crop up in ML diagrams. That imbalance leaves the numerical and semantic rows of Table 2 on shakier statistical ground, so those figures are best read as indicative rather than final. Firmer quantitative claims would need a larger corpus, sampled evenly across error types and across domains such as biochemistry and physics. One further gap: the Advocate-Critic loop is specified at the design level but not yet validated empirically, and how it moves TPR, OR, and LocAcc relative to single-agent prompting is left for future work.

5. Conclusion

Reliable scientific video generation, this work argues, depends on catching structural and topological errors in figures before anything is synthesised. In the diagnostic evaluation here, current VLMs – proprietary and open-weight alike – fail as autonomous verifiers in three recurring ways: Knowledge Contamination, Spatial Insensitivity, and the Chain-of-Thought Paradox. Across every configuration we tried, a model either rejects correct figures or misses injected ones, with worst-case rates touching 0.86 in both directions. General-purpose VLM prompting, in short, is not enough for production-level verification of scientific multimedia. The Agentic Director Pipeline answers that with source-grounded textual checking, figure-type-specific structural schemas, adversarial verification roles, and explicit escalation – making Keyframe Verification an atomic safety gate for PDF-to-video generation. The corpus is still limited in size and coverage, but it gives a foundation to scale evaluation toward verifiers built specifically for scientific multimedia.

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A. Supplementary Materials

A.1. Prompts for Models

Zero-shot prompt (Claude Sonnet 4):

Look at this scientific figure. Does it contain any factual, structural, or labeling errors?

Answer strictly PASS if the figure is fully correct, or FAIL if there is ANY error. If FAIL, explain what exactly is wrong in 1-2 sentences.

Chain-of-Thought prompt (Claude Sonnet 4):

`<role>You are an expert scientific peer-reviewer specializing in validating figures for academic publications.</role>`

`<task>Examine the provided scientific figure and determine whether it is fully accurate. Perform your analysis inside <thinking> tags following these steps:`

1. IDENTIFY: What type of figure is this?
2. READ: List every text label, number, and annotation visible.
3. TRACE: If diagram, trace data flow. If chart, read axes and values.
4. VALIDATE: Compare every element against your knowledge.
5. DECIDE: Is there ANY error?`</task>`

`<output_format>After your analysis, output:
<verdict>PASS</verdict> or <verdict>FAIL</verdict>`

If FAIL, state:

- Error type: [structural / numerical / labeling]
- What is wrong: [description]
- What it should be: [correct version]`</output_format>`

A.2. Detailed Error Taxonomy

Structural errors are violations of a figure’s topology, and they take five forms. In block displacement a component turns up in the wrong region of the graph. In block swapping two adjacent components trade places. Component deletion removes an element the architecture needs; its mirror image, the superfluous component, adds one that does not belong. The fifth, incorrect connection direction, reverses an arrow so that it runs against the intended flow of data.

Numerical errors leave the topology intact and corrupt the quantities instead. There are four kinds: an incorrect dimensionality, where an embedding or hidden size no longer matches the model’s specification; a wrong count of layers or attention heads; a wrong vocabulary size; and numerical parameters that have been swapped between two different models.

Semantic errors keep both the topology and the numbers but break the textual grounding. Three kinds occur: an incorrect component name, such as LayerNorm written where the figure should read RMSNorm; incorrect attribution, where one model’s results are labelled as another’s; and an incorrect abbreviation, GQA standing in for MHA or the reverse.

A.3. Examples of Modified Figures

This section collects visual examples from the diagnostic corpus, placing each original figure next to its adversarially modified counterpart across the structural, numerical, and semantic error classes.

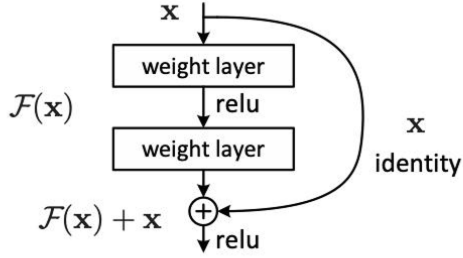


Figure 2. Residual learning: a building block.

(a) Original block.

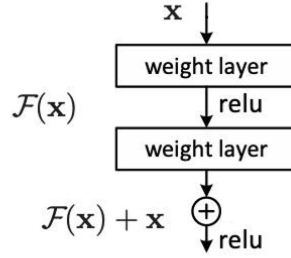


Figure 2. Residual learning: a building block.

(b) Modified block.

Figure 2. A **structural error (component deletion)**. The identity shortcut – the skip connection – of the canonical ResNet (He et al., 2016) block has been removed entirely. VLMs tend to miss the broken topological link and hallucinate the connection as still present.

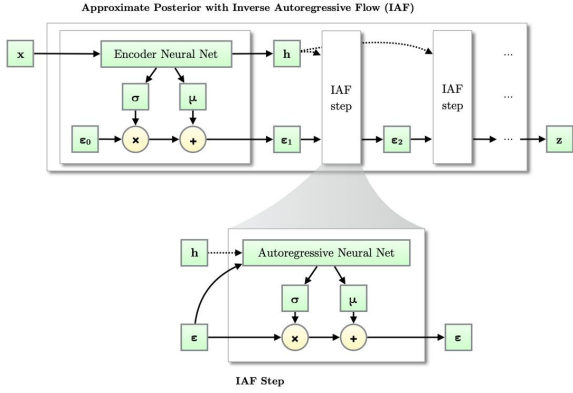


Figure 3.1: Like other normalizing flows, drawing samples from an approximate posterior with Inverse Autoregressive Flow (IAF) (Kingma et al., 2016) starts with a distribution with tractable density, such as a Gaussian with diagonal covariance, followed by a chain of nonlinear invertible transformations of z , each with a simple Jacobian determinant. The final iterate has a flexible distribution.

(a) Original IAF diagram.

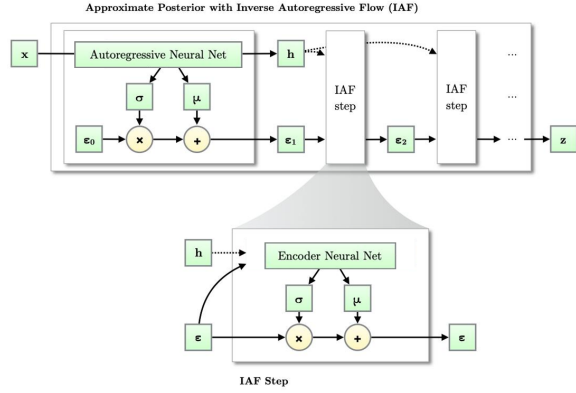
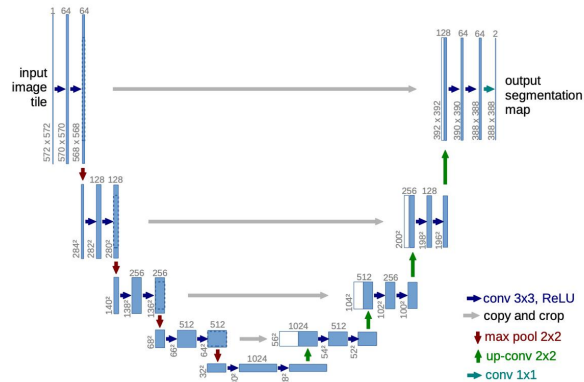


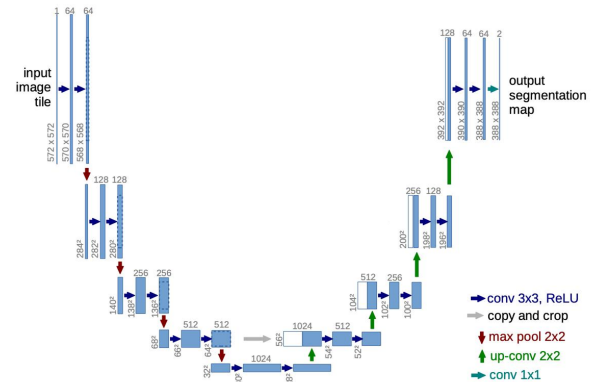
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(b) Modified diagram.

Figure 3. A **structural error (block swapping)**. The “Encoder Neural Net” and “Autoregressive Neural Net” blocks have changed places. The labels are all correct, yet the directed flow through the graph is now wrong.

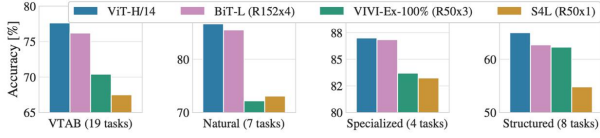


(a) Original architecture.

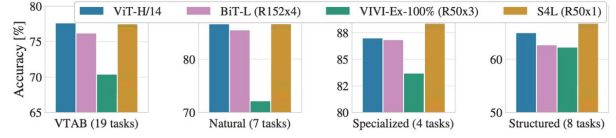


(b) Modified architecture.

Figure 4. A **structural error (component deletion)**. The U-Net’s signature cross-layer “copy and crop” connections have been deleted, which effectively collapses it into an ordinary autoencoder.

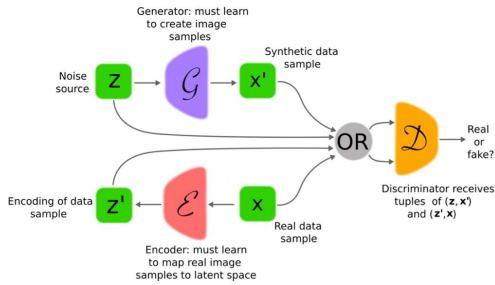

 Figure 2: Breakdown of VTAB performance in *Natural*, *Specialized*, and *Structured* task groups.

(a) Original benchmark chart.

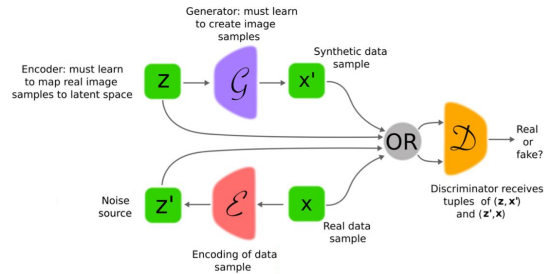

 Figure 2: Breakdown of VTAB performance in *Natural*, *Specialized*, and *Structured* task groups.

(b) Modified chart.

Figure 5. A **numerical error**. In the “Natural” task group the gold S4L bar has been inflated to sit level with SOTA performance, which contradicts both the visual scale and the established results.


 Fig. 4. The ALIBIGAN structure [20], [19] consists of three networks. One of these serves as a discriminator, another maps the noise vectors from latent space to image space (decoder, depicted as a generator $\mathcal{G}$ in the figure), with the final network (encoder, depicted as $\mathcal{E}$) mapping from image space to latent space.

(a) Original structure.


 Fig. 4. The ALIBIGAN structure [20], [19] consists of three networks. One of these serves as a discriminator, another maps the noise vectors from latent space to image space (decoder, depicted as a generator $\mathcal{G}$ in the figure), with the final network (encoder, depicted as $\mathcal{E}$) mapping from image space to latent space.

(b) Modified structure.

Figure 6. A **semantic error (swapped attribution)**. The text descriptions of the Generator and the Encoder have been exchanged. Topology is untouched, but the semantic grounding is now inverted.